

Education

B.S. Computer Science: Game Development Concentration | North Carolina State University, Raleigh, NC
Minors in Creative Writing & English | Summa Cum Laude | University Honors Program | Dean’s List
Summer Study Abroad | University of Oxford, Oxford, UK

May 2026
GPA: 3.9/4.0
Jun 2024 – Jul 2024

Relevant Coursework: Advanced Game Projects, Advanced Fiction Writing, Game Engine Foundations, Computer Graphics, Introduction to Artificial Intelligence, Communication for Engineering & Technology, Software Product Management

Skills

Languages: C++, C, C#, Java, JavaScript, Python, HTML, CSS
Frameworks & Libraries: SDL3, ZeroMQ, JUnit, Bootstrap, React, Tailwind CSS, Spring, REST
Game Engines: Unreal Engine 5 (C++, Blueprints), Godot (GDScript, C#), Ren’Py, Twine
Tools: GitHub, GIT, Visual Studio, Eclipse, Perforce, Jenkins, Maven, Adobe Suite, Notion, Obsidian, Figma, Blender

Work Experience

Gameplay Programmer | Frontline Interactive Jun 2026 – Present

Raleigh, NC | Start-up games studio creating gameplay first serious games, contract with Miller Fellowship at NC State

- Principal gameplay programmer for Firechess, a serious game developed as part of ongoing research into a new wildfire response training program for the US National Guard
- Served in various roles on a small studio team; focusing on mechanic design, C++ programming, production, and narrative

Stage Manager | NC State University Theatre Sep 2022 – Present

Raleigh, NC | Shows: *The Prom* ('26), *Dial M for Murder* ('25), *Mr. Burns: A Post-Electric Play* ('24), and 50+ event operations productions

- Managed cross functional teams of 60+ company members across technical and creative disciplines, ensuring production timelines were kept and milestones were hit on schedule

Co-Author, Published Research Paper | NC State University Nov 2024 – Feb 2025

"Advancing Software Product Management Education: Insights from an Industry Survey" — *Foundations of Software Engineering Conference 2025*

- Collaborated with a 6-person research team to categorize industry survey findings into 21 key competencies and 10 defining factors that characterize strong Product Managers
- Gathered, analyzed, and synthesized survey data from 122 industry professionals across 24+ companies

Help Desk Technician | Collegiate School May 2019 – Aug 2019

Henrico, VA | Summer IT support for school-wide technology operations

Project Experience

Firechess | Strategy Game — Unreal Engine 5, C++, Blueprints

Team project 5-person team: Hexagon based strategy game simulating wildfire-fighting tactics in a turn-based format

- Led design and programming of the base Unit class, serving as the foundation for all unit game pieces
- Implemented 5 unique unit types, defining their distinct abilities and tactical roles within the game's strategic framework
- Wrote 15 original narrative scenarios and engineered the alert system that delivers them, entwining narrative with gameplay

The Deep End | Adventure Game — Godot Engine, Triangle Game Jam 2026 Submission

Team project 5-person team: Designed, built, play tested, and shipped within one week — itch.io release

- Implemented the objective spawning system, designing the logic that governs goal placement and progression
- Sole programmer for enemy AI, combat, and procedural enemy spawning system, driving the game’s tension
- Built a dialogue system and accompanying UI from scratch that adapts to player actions and game world changes

2D Game Engine | Custom Game Engine — C++, SDL3

Team project 3-person team: Comprehensive game engine built from scratch; networking implemented via pub-sub and request-reply patterns

- Led implementation of the engine's networking functionality, ensuring expected operation across a client-server connection
- Implemented support for death, input, and collision events within the networked architecture
- Programmed a minigame at each development milestone to validate and test the engine’s toolset in a live context

Honors & Certifications

Epic Games: Game Design Professional Certificate	In Progress
Bitzter Theatre Award NC State University Theatre	Apr 2025
Alpha Psi Omega — Psi Kappa Cast National Theatre Honors Society Member	Mar 2024